

Coursework 2 – Topic Proposal

Predictive Model for Malicious Network Activity Detection

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1. Introduction

The goal of this project is to create a predictive model using machine learning capabilities to detect potentially malicious network activities or unauthorized access attempts such as PortScanning and Denial of Service (DoS). By focusing on specific network behaviors captured in the CICIDS2017 dataset, the project will apply different machine learning models to classify and predict certain type of activities.

Cybersecurity attacks can cause financial losses as well as major disruptions to their victims. For this particular reason, implementing a proactive approach to such risks is essential to address vulnerabilities and preventing potential breaches. Leveraging different technologies such as Hadoop, PySpark and MLlib, and focusing on two main network attack types – PortScanning and DoS -, this project aims at developing a binary classifier that will be able to predict whether a given activity can be deemed as malicious.

Although the scope of the project is limited to detecting only two particular network attacks, the study is a good starting point for the creation of a much more robust and comprehensive Intrusion Detection Systems (IDSs) and Intrusion Prevention Systems (IPSs)¹. Hence, the project can be scaled to a bigger dataset in order to study the behavior of other types of suspicious activities that can take place in a network.

2. Data

2.1 Data Collection and summary

The dataset CICIDS2017 was retrieved from Kaggle² and was specifically chosen as it captures different network behaviors, including both benign and malicious activities. The data was publicly available in the format of eight different .csv files that gather the network traffic behavior over the span of five days.

The dataset is composed of 79 variables or features that record different network traffic activity attributes such as timestamps, logs, traffics, length and frequency of packets transmitted. Thanks to the numerous variables available in the dataset, as well as the presence of both benign and malicious activity, patterns of behavior can be analyzed and mapped for the analysis. Moreover, the dataset includes data describing the patterns of a range of network attacks such as: Brute Force FTP, Brute Force SSH, DoS, Heartbleed, Web Attack, Infiltration, Botnet and DDoS. Nevertheless, for the purpose of this project, we will focus on only two network attacks: DoS and PortScan.

¹ <https://www.kaggle.com/datasets/cicdataset/cicids2017/data>

² <https://www.kaggle.com/datasets/cicdataset/cicids2017/data>

2.2 Data cleaning and preprocessing

2.2.1 Data Uploading

The .csv files were downloaded from public repository and uploaded to the university cluster's local file system 'lena'. Files were saved into a new directory called 'Final_BDA' (path to the directory '/home/asell001/Final_BDA')

```
asell001@lena:~$ hadoop fs -copyFromLocal ./Final_BDA
asell001@lena:~$ hadoop fs -ls
Found 8 items
drwxr-xr-x  - asell001 users          0 2024-03-12 09:35 .sparkStaging
drwxr-xr-x  - asell001 users          0 2024-03-14 13:55 Final_BDA
drwxr-xr-x  - asell001 users          0 2023-12-30 13:29 Topic2bis
drwxr-xr-x  - asell001 users          0 2024-01-02 17:26 Topic3bis
-rw-r--r--  3 asell001 users 12078583 2024-01-02 17:48 books.txt
drwxr-xr-x  - asell001 users          0 2024-01-06 13:06 docs
drwxr-xr-x  - asell001 users          0 2024-01-07 17:07 exercisel
drwxr-xr-x  - asell001 users          0 2024-03-11 09:55 output_spark
```

The files contained in the new directory were then uploaded to HDFS:

```
asell001@lena:~$ hadoop fs -ls Final_BDA
Found 8 items
-rw-r--r--  3 asell001 users  77123859 2024-03-14 13:55 Final_BDA/1_Friday-WorkingHours-Afternoon-DDos.pcap_ISCX 2.csv
-rw-r--r--  3 asell001 users  76906168 2024-03-14 13:55 Final_BDA/2_Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv
-rw-r--r--  3 asell001 users  58316725 2024-03-14 13:55 Final_BDA/3_Friday-WorkingHours-Morning.pcap_ISCX.csv
-rw-r--r--  3 asell001 users 176927918 2024-03-14 13:55 Final_BDA/4_Monday-WorkingHours.pcap_ISCX.csv
-rw-r--r--  3 asell001 users  83102436 2024-03-14 13:55 Final_BDA/5_Thursday-WorkingHours-Afternoon-Infiltration.pcap_ISCX.csv
-rw-r--r--  3 asell001 users  52023263 2024-03-14 13:55 Final_BDA/6_Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX 2.csv
-rw-r--r--  3 asell001 users 135078995 2024-03-14 13:55 Final_BDA/7_Tuesday-WorkingHours.pcap_ISCX.csv
-rw-r--r--  3 asell001 users 225166395 2024-03-14 13:55 Final_BDA/8_Wednesday-workingHours.pcap_ISCX.csv
```

The files were then loaded to Jupyter Notebook and merged into a single DataFrame:

```
# Load and merge dataset and create DataFrame
file_paths = 'hdfs:///user/asell001/Final_BDA/*.csv' # Pattern to match all CSV files in the directory
df = spark.read.option("header", "true").option("inferSchema", "true").csv(file_paths)
```

2.2.2 Data cleaning

While preparing the DataFrame for analysis, one of the first issue encountered was with the presence of leading whitespace (' ') before the name of some of the variables in the schema. This inconsistency can cause problems in the analysis as the exact column name is needed in the code; not knowing whether there's a whitespace before a variable's name can be problematic. This issues was addressed in the following way:

```
# Get the current column names from the DataFrame
current_columns = df.columns

# Create a dictionary mapping current column names to trimmed column names
renamed_columns = {col_name: col_name.strip() for col_name in current_columns}

# Use selectExpr to select and rename columns in one go
df_clean = df.selectExpr(*(f"{col}" as `{new_col}`" for col, new_col in renamed_columns.items()))

# Verify the new column names
df_clean.printSchema()
```

2.2.3 Data preprocessing

As previously mentioned, this project will be selecting only those variables that are most relevant to describe and capture behaviors associated with DoS and PortScanning attacks. The choice of these two particular network attacks was dictated by the number of available records in the dataset. Indeed, a closer look at 'Label' column shows that the majority of suspicious usage gathered in the dataset was labeled either as DoS or PortScan.

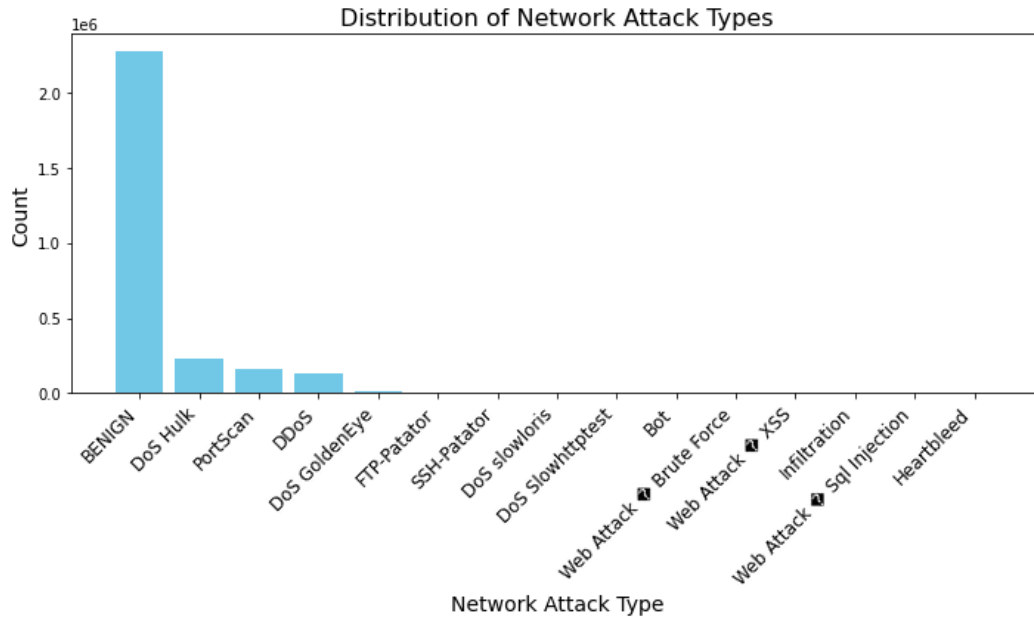
```
# Group the data by the 'Label' column and count the occurrences
label_counts = df_clean.groupBy(col("Label")).count()

# Sort the counts in descending order
label_counts = label_counts.orderBy(col("count").desc())

# Show the result
label_counts.show(truncate=False)
```

Label	count
BENIGN	2273097
DoS Hulk	231073
PortScan	158930
DDoS	128027
DoS GoldenEye	10293
FTP-Patator	7938
SSH-Patator	5897
DoS slowloris	5796
DoS Slowhttptest	5499
Bot	1966
Web Attack 0 Brute Force	1507
Web Attack 0 XSS	652
Infiltration	36
Web Attack 0 Sql Injection	21
Heartbleed	11

Given the abundance of observations related to PortScan and general DoS network attacks, it was logic to use these attacks as focus for the analysis.



After having established the focus of the analysis, the next step is to understand what are the most relevant variables to take into considerations when it comes to mapping PortScan and DoS behavior. In order to do so, it is necessary to firstly understand the Tactics, Techniques, and Procedures (TTPs) of each different attack.

PortScanning is a technique utilized to scan a network with the aim of identifying open ports on a target host. After having mapped any potential exposed service, bad actors can go on and illegitimately infiltrate in the victim's network. PortScanning is used as initial technique to gather intelligence on potential network vulnerabilities. DoS attacks, on the other hand, aim at overwhelming a network by flooding it with numerous requests. Due to the large amount of requests received, the server is unable to process them ultimately causing denial of service (DoS) also to legitimate requests.

Based on these definitions, it is correct to say that certain indicators are much more meaningful for the study of specific types of network attacks. For example, features such as 'Flow Duration', 'Total Fwd Packets', 'Total Backward Packets' are extremely relevant for DoS attacks as they provide information on the volume of traffic. Moreover, 'Fwd Packet Length Max', 'Fwd Packet Length Min', 'Bwd Packet Length Max', 'Bwd Packet Length Min' and 'Flow Bytes/s' are also central in detecting DoS attacks as the modus operandi of this particular attack entails suspicious package size and frequency³⁴.

Variables such as 'Flow Duration', 'Total Fwd Packets' and 'Total Backward Packets' were also selected to map PortScanning activity, however, considering the specific TTPs of such network attack, additional features such as 'Destination Port', 'Fwd IAT Total', 'Bwd IAT Total' (Inter-Arrival Times) are to be included in the analysis⁵.

³ <https://www.fortinet.com/resources/cyberglossary/dos-vs-ddos>

⁴ <https://www.cloudflare.com/en-gb/learning/ddos/glossary/denial-of-service/>

⁵ <https://www.avast.com/business/resources/what-is-port-scanning#mac>

3. Prediction Model

In order to create a more accurate prediction model, two different DataFrames for each different attack type were created. The aim of this step was to isolate the relevant variables for each different attack from the broader dataset. The code snippet below show the creation of the two DataFrames:

```
#Create DataFrame including variables that can predict DoS attacks

dos_columns = [
    "Flow Duration", "Total Fwd Packets", "Total Backward Packets",
    "Flow Bytes/s", "Flow Packets/s", "Fwd Packet Length Max", "Fwd Packet Length Min",
    "Fwd Packet Length Mean", "Fwd Packet Length Std", "Bwd Packet Length Max", "Bwd Packet Length Min",
    "Bwd Packet Length Mean", "Bwd Packet Length Std", "SYN Flag Count", "PSH Flag Count",
    "ACK Flag Count", "Label"
]

dos_df = final_df.select(*dos_columns)
```

```
#Create DataFrame including variables that can predict PortScan attacks

portscan_columns = [
    "Destination Port", "Flow Duration", "Total Fwd Packets", "Total Backward Packets",
    "Fwd IAT Total", "Bwd IAT Total", "Label"
]
```

Each different DataFrame was then encoded into two distinct categories: benign activities encoded as '0' and all other types of attacks encoded as '1'. The binary encoding facilitates the classification task, enabling the model to effectively differentiate between normal and malicious behavior.

```
# Create a new column 'BinaryLabel' where 'BENIGN' is encoded as 0 and all types of attacks as 1

dos_df_encoded = dos_df_ready_for_ml.withColumn("LabelEncoded", when(col("Label") == "BENIGN", 0).otherwise(1))
```

```
# Create a new column 'BinaryLabel' where 'BENIGN' is encoded as 0 and all types of attacks as 1

portscan_df_encoded = portscan_df.withColumn("LabelEncoded", when(col("Label") == "BENIGN", 0).otherwise(1))
```

For the purpose of this project, two different machine learning models were applied to the different DataFrames: logistic regression and random forest models. Their creation involved critical steps including assembly of feature vectors, scaling data and diving the data into test and training as well as its evaluation. Detailed information on the pipeline construction and relevant code snippet can be found on the attached Jupyter notebook file.

In the table below the area under the ROC scores are gathered for each different dataframe and model:

	Logistic Regression	Random Forest
DoS DataFrame	0.5	0.99
PortScan DataFrame	0.97	0.99

Based on the above scores we can draw the following conclusions:

- The random forest model outperformed the logistic regression model in both attack detection scenarios. This is most likely thanks to the model's ability to handle non-linear features
- Logistic regression model while failing in detecting DoS attacks (0.5 score), showed efficacy in identifying PortScan activities suggesting good performance in those scenarios where the relationship between variables is more linear

4. Summary & Conclusion

The goal of this project is to create a predictive model using machine learning capabilities to detect potentially malicious network activities or unauthorized access attempts by focusing on two main network attack types: PortScanning and DoS. By leveraging the data available in CICIDS2017 dataset, the project applied different machine learning models to classify and predict different malicious network activity.

Throughout the project, various challenges were encountered, including the identification of relevant features and the presence of whitespace in column names. These issues were addressed through cleaning process and feature engineering techniques that entailed narrowing down the dataset to variables most indicative of the cyberattacks under scrutiny. Moreover, the project compared the performance of two different machine learning models: logistic regression and random forest.

The project's findings highlight several key points: firstly, the critical importance of feature selection for model performance. Indeed, focusing on variables directly associated with the cyberattacks of interest and isolate the right features in the dataframe can enhance model's accuracy. Secondly, the effectiveness of random forest over regression model in cybersecurity contexts; this can be attributed to random forest model's ability to identify intricate attack patterns within network traffic data.

In conclusion, the project has demonstrated the potential that machine learning models have on cybersecurity threat detection. Continuous refinement of machine learning techniques and data gathering is essential to keep pace with the evolving landscape of cyber threats.

5. Resources

Canadian Institute for Cybersecurity. (n.d.). CICIDS2017. Kaggle. Retrieved from <https://www.kaggle.com/datasets/cicdataset/cicids2017/data>

Avast. (n.d.). What is Port Scanning? Avast Business Resources. Retrieved from <https://www.avast.com/business/resources/what-is-port-scanning#mac>

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